

University Attendance System Suitable Data Structure for Recording Student Attendance

Presented by:

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Problem Statement

- ▶ **Scenario**
- ▶ A university records student attendance **sequentially** during every class hour.
- ▶ Each student's attendance is entered one after another.
- ▶ The system should allow **fast insertion** of new attendance records.
- ▶ **Requirement**
- ▶ Choose a suitable data structure that provides efficient insertion.

Recommended Data Structure

- ▶ **Data Structure: Linked List**
- ▶ **Why Linked List?**
- ▶ Stores attendance records in sequence.
- ▶ New records can be inserted quickly.
- ▶ Memory is allocated dynamically.
- ▶ No need to shift existing records during insertion.

Advantages

- ▶ Fast insertion of attendance records.
- ▶ Flexible size.
- ▶ Efficient memory utilization.
- ▶ Easy to add new students during class.

Conclusion

- ▶ **Recommended Data Structure: Linked List**
- ▶ It is ideal because attendance is recorded **sequentially**.
- ▶ Provides **efficient insertion ($O(1)$)** when using a tail pointer.
- ▶ Suitable for dynamic attendance systems where records are continuously added.

THANK YOU